

6	(a) energy converted from chemical to electrical per unit charge	B1	[1]
	(b) (i) current = $E/(R + r)$	C1	
	$= 6.0/(16 + 0.5)$		
	$= 0.36 \text{ (0.364) A}$	A1	[2]
	(ii) terminal p.d. = $(0.36 \times 16) = 5.8 \text{ V}$ or $(6 - 0.36 \times 0.5)$		
	$= 5.8 \text{ V}$	A1	[1]
	(c) (i) use of $R = \rho l/A$ or proportionality with length and inverse proportionality with area or d^2	C1	
	$d/2$ and $l/2$ gives resistance of $Z = 2R_Y = 24 (\Omega)$	C1	
	R = resistance of parallel combination = $[1/24 + 1/12]^{-1}$		
	$= 8(.0) (\Omega)$	A1	[3]
	(ii) resistance of circuit less therefore current larger	B1	
	lost volts greater therefore terminal p.d. less	B1	[2]
	(d) power = $I^2 R$ or VI or V^2/R	C1	
	current in second circuit (= $6.0/12.5$) = 0.48 (A)	B1	
	ratio = $[(0.36)^2 \times 16] / [(0.48)^2 \times 12] = 0.75$ [0.77 if full s.f. used]	B1	[3]

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2015	9702	21